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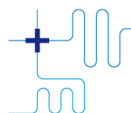
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ABSTRACT

In vertical-channel transistors used as access transistors in high-density dynamic random-access memory, the body becomes electrically floating, leading to a floating-body effect (FBE) that degrades data retention. Most previous studies have focused on reducing gate-induced drain leakage (GIDL) at the storage node junction to improve “1” state retention, while leaving the “0” state retention unaddressed. This work introduces a three-layer work function (WF) gate structure that symmetrically suppresses the FBE for both states. Low-WF layers are positioned at both ends of the channel to symmetrically suppress GIDL, whereas a high-WF layer in the center maintains channel depletion and thereby prevent retention degradation. Mixed-mode TCAD simulations of single-, dual-, and three-layer WF gates under both dynamic and static retention conditions demonstrate that the proposed design effectively suppresses the FBE in both the “1” and “0” states.

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Continuous scaling of one-transistor-one-capacitor (1T1C) dynamic random-access memory (DRAM) cells leads to severe degradation in retention time and parasitic coupling.^{1–5} Vertical or trench-type capacitors have been introduced to maintain sufficient storage capacitance, but these structures still rely on planar-channel access transistors and cannot reduce the footprint below $6F^2$, where F denotes the minimum lithographic feature size.^{6–9}

To overcome this limitation, vertical-channel transistors (VCTs) have been proposed as access transistors, forming the channel along a cylindrical pillar with vertically stacked bit line (BL), word line (WL), and capacitor, thereby realizing a $4F^2$ cell footprint.^{10,11} Although this structure improves scalability, the silicon body becomes electrically floating due to surrounding oxides. As a result, holes generated by gate-induced drain leakage (GIDL) through band-to-band tunneling (BTBT) accumulate in the body, increasing the body potential. This floating-body effect (FBE) induces threshold-voltage instability, read disturbance, and retention degradation, posing major reliability challenges for DRAM applications.^{12,13}

Thinning the body to achieve full depletion can mitigate the FBE, but process constraints limit its practicality.^{14,15} A feasible

approach is to employ junction and gate work function engineering to suppress GIDL-induced hole generation.^{9,16} However, most previous studies have focused exclusively on “1” state retention degradation, which is dominated by GIDL at the storage node (SN) junction, while neglecting “0” state retention degradation dominated by GIDL at the BL junction.^{12,14,17–19} In particular, in VCTs, applying a dual-WF metal gate designed for conventional buried-channel array transistors (BCATs) improves “1” state retention but degrades “0” state retention, because while GIDL is suppressed at the SN junction, it increases at the BL junction.²⁰ Therefore, a symmetric suppression strategy that mitigates FBE-induced retention degradation in both states is required.

In this work, a three-layer WF gate is proposed for the VCT DRAM, as illustrated in Fig. 1(a). Low-WF layers (M1 and M3) are placed at both ends of the channel to suppress GIDL, while a central high-WF layer (M2) maintains channel depletion, as shown in Fig. 1(b). Comparative analyses of single-, dual-, and three-layer WF gates under dynamic and static retention conditions for both “1” and “0” states demonstrate that the proposed design effectively suppresses the FBE in all states.

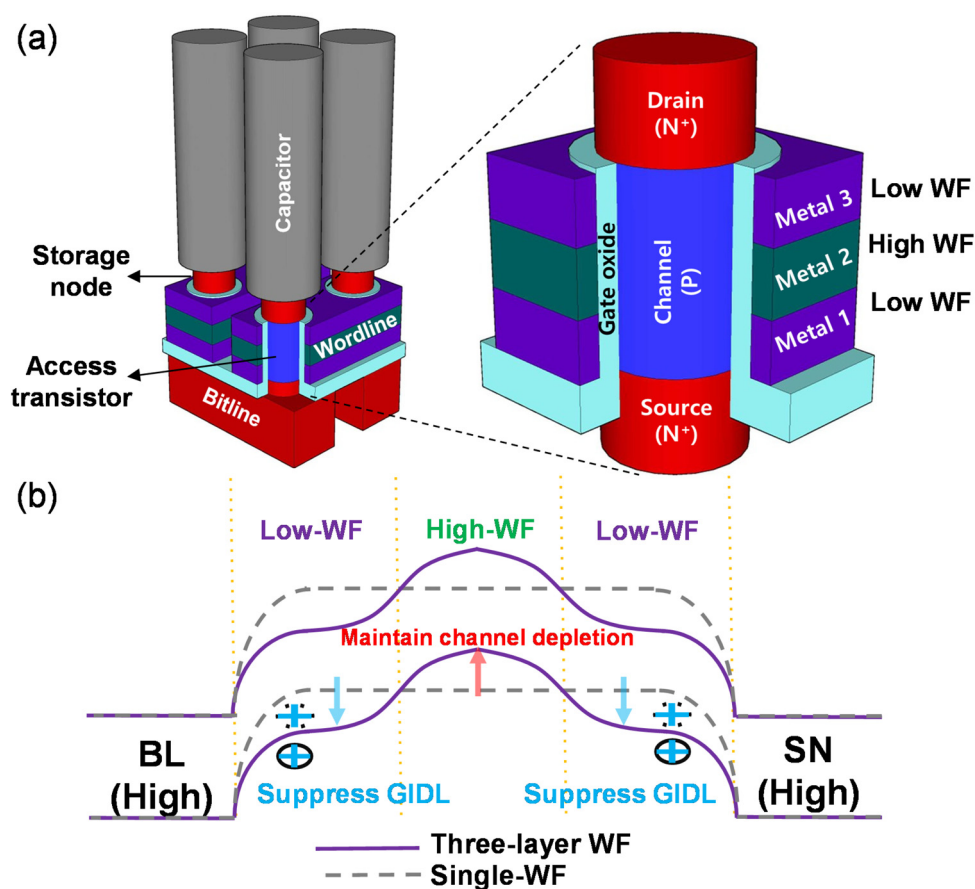


FIG. 1. Three-layer work function (WF) gate designed to suppress floating-body effects in vertical-channel transistor (VCT) DRAM. (a) Three-dimensional schematic and (b) energy band diagram.

A VCT functioning as a DRAM access transistor was implemented using Synopsys Sentaurus TCAD.²¹ The baseline device structure was adopted from a previously reported pillar-type VCT, which has body thickness, gate oxide thickness, and gate length of 400, 20,

and 400 nm.²² The physical models and parameters were first calibrated against experimental data from the reference pillar-type VCT, as confirmed by the close agreement of transfer characteristics (I_D - V_G) shown in Fig. 2(a). Based on the calibrated models and parameters, the

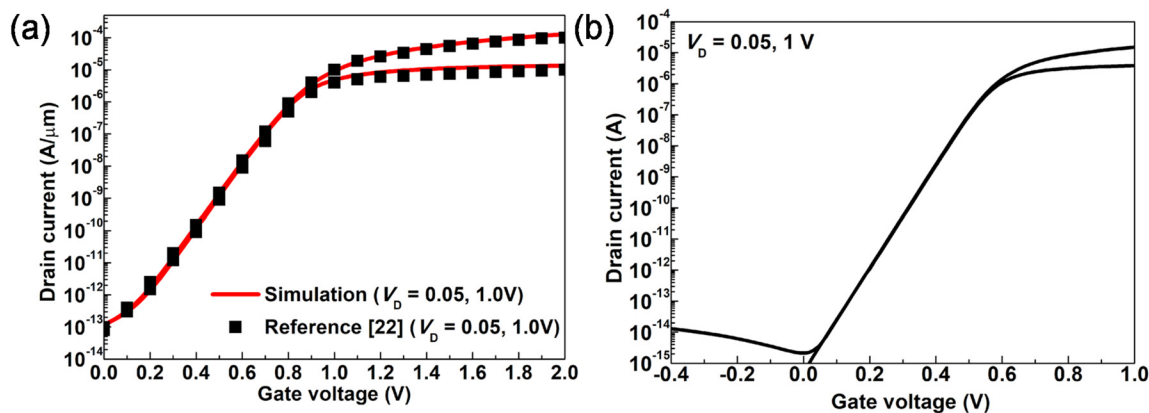


FIG. 2. (a) Transfer characteristics (I_D - V_G) of the reference VCT used for TCAD calibration. (b) Transfer characteristics (I_D - V_G) of the rescaled VCT used in floating-body effect analysis.

device geometry was rescaled to a body thickness of 12 nm, a gate oxide thickness of 1 nm, and a gate length of 60 nm to investigate FBE in a scaled access transistor, where GIDL and BTBT become the dominant sources of hole generation and the primary contributors to FBE. The silicon body was doped to a concentration of $1 \times 10^{17} \text{ cm}^{-3}$, while the source/drain were doped to $1 \times 10^{20} \text{ cm}^{-3}$ using Gaussian doping profiles. The transfer characteristics of the rescaled VCT structure are shown in Fig. 2(b).

The proposed three-layer WF gate can be fabricated by modifying the gate-stack formation step after pillar formation in a VCT fabrication process. In this approach, three metal layers (M1, M2, and M3)

are sequentially deposited from bottom to top, where M1 and M3 have low WFs, whereas M2 has a high WF. In this approach, each metal layer should not fully cover the vertical channel but instead be deposited in a bottom-up manner. Therefore, a deposition technique with limited step coverage, such as evaporation, should be employed. A mixed-mode framework was employed to simulate the 1T1C operation, where the access transistor was connected to a 10 fF storage capacitor and a 100 fF BL capacitor.

The physical models enabled in the simulation include nonlocal BTBT, Shockley–Read–Hall (SRH) recombination, phonon-limited mobility (PhuMob), and high-field saturation, which together capture

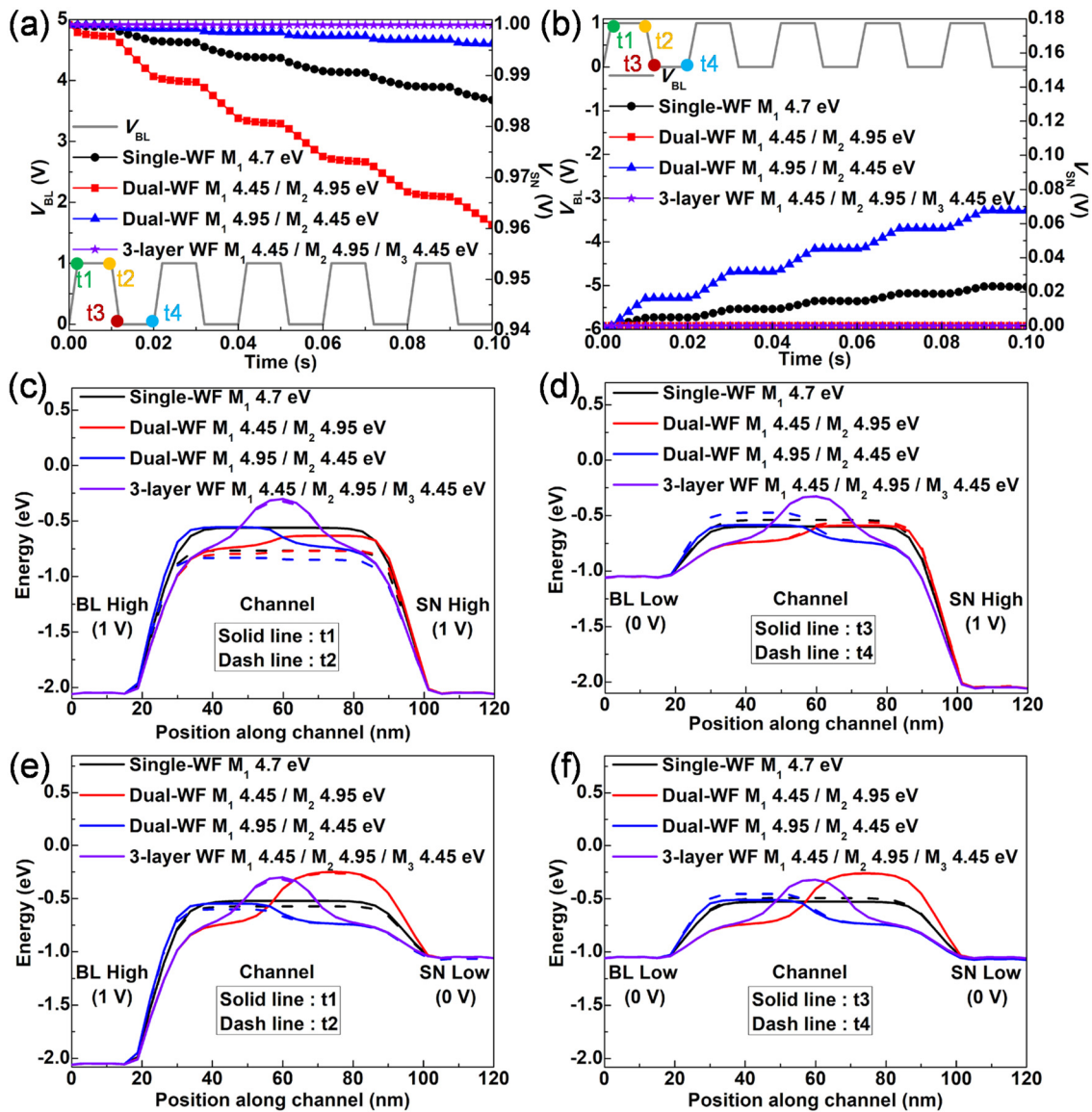


FIG. 3. Dynamic retention characteristics of single-, dual-, and three-layer WF gate structures. (a) “1” state and (b) “0” state dynamic retention behavior showing the transient response of V_{SN} in accordance with BL toggling. Valence-band energy profiles along the channel during the (c) BL-high period (t1–t2) and (d) BL-low period (t3–t4) for the “1” state. Valence-band energy profiles along the channel during the (e) BL-high interval (t1–t2) and (f) BL-low interval (t3–t4) for the “0” state.

the dominant transport, carrier generation, and recombination mechanisms governing GIDL and FBE in scaled VCT.

Specifically, nonlocal BTBT is required to accurately describe hole generation under the strong junction electric fields associated with GIDL, which constitutes the primary physical origin of FBEs. The subsequent accumulation and decay of holes in the body are governed by SRH recombination, determining the temporal evolution of body potential behavior. PhuMob provides a physically consistent description of channel conduction in response to body potential modulation, while high-field velocity saturation prevents unphysical carrier acceleration and accurately captures carrier transport under the strong electric fields inherent to scaled VCT geometries.

Dynamic retention refers to the data retention behavior when the WL is turned off while the BL is toggling. As the BL voltage (V_{BL}) toggles, holes are generated and stored in the floating body, while the storage capacitor loses charge and the SN voltage (V_{SN}) shifts.¹⁹ Figure 3 shows the simulated dynamic retention characteristics of the single-, dual-, and three-layer WF gates for both the “1” and “0” states. Figures 3(a) and 3(b) show the transient response of V_{SN} under periodic V_{BL} toggling for the “1” and “0” states, respectively. Figures 3(c)–3(f) depict the corresponding valence-band energy profiles along the vertical channel at each time step (t_1 – t_4) indicated in Figs. 3(a) and 3(b).

When the SN stores a “1” state, GIDL occurs at both the SN and BL junctions during BL-high period (t_1 – t_2), injecting holes into the body and raising the body potential, as shown in Fig. 3(c). During BL-low period (t_3 – t_4), as shown in Fig. 3(d), electrons are injected from the BL to the SN, decreasing V_{SN} . FBE accelerates this V_{SN} degradation because the stored holes reduce the potential barrier between the BL and the body. Consequently, repeated BL toggling gradually lowers V_{SN} .

For the dual-WF gate, when the SN-side metal (M2) has a low WF (4.45 eV) and the BL-side metal (M1) has a high WF (4.95 eV), the weaker electric field at the SN junction suppresses GIDL, and the higher potential barrier between the BL and body blocks electron injection, yielding improved retention compared to the single-WF gate. Conversely, when M2 uses a high WF and M1 a low WF, GIDL at the

SN junction and electron injection from the BL toward the SN are enhanced, producing the most severe V_{SN} degradation. The three-layer WF gate shows the smallest V_{SN} degradation, as the low WF at M3 and M1 suppresses GIDL at both the SN and BL junctions, while the high WF at M2 maintains channel depletion and prevents electron injection from the BL toward the SN. This comparison indicates that, unlike the dual-WF gate whose retention behavior strongly depends on the WF assigned to each junction, the three-layer WF gate provides a symmetric electrostatic environment that suppresses FBE-induced retention degradation in a state-independent manner.

When the SN stores a “0” state, GIDL occurs at the BL junction during BL-high period (t_1 – t_2), injecting holes into the body, as shown in Fig. 3(e). Simultaneously, electrons are injected from the SN to the BL, increasing V_{SN} . Here, FBE accelerates the V_{SN} rise because stored holes lower the potential barrier between the SN and the body. During BL-low period (t_3 – t_4), a slight drop in V_{SN} occurs due to electron backflow from BL to SN, as shown in Fig. 3(f).

Unlike the “1” state, the dual-WF configuration with a high WF at M2 and low WF at M1 exhibits improved retention compared to the single-WF gate, since GIDL and electron injection from the SN to the BL are suppressed. However, the opposite low WF at M2 and high WF at M1 leads to the poorest retention. The three-layer WF gate achieves good retention performance because the low WF at M3 and M1 suppresses GIDL and the high WF at M2 prevents electron injections from the SN toward the BL.

Figure 4(a) shows the fluctuation of V_{SN} after ten BL toggling cycles, clearly demonstrating the superior dynamic retention of the three-layer WF gate for both the “1” and “0” states. Figure 4(b) presents the body potential increase during the BL-high period (t_1 – t_2), further confirming that the three-layer WF gate effectively suppresses the FBE under both states.

Static retention refers to the data retention behavior when the WL is turned off and the BL is precharged to $V_{DD}/2$ (0.5 V). Figure 5 presents the simulated static retention characteristics of the single-WF, dual-WF, and three-layer WF gates for both the “1” and “0” states. Figures 5(a) and 5(b) show V_{SN} over time for the “1” and “0” states,

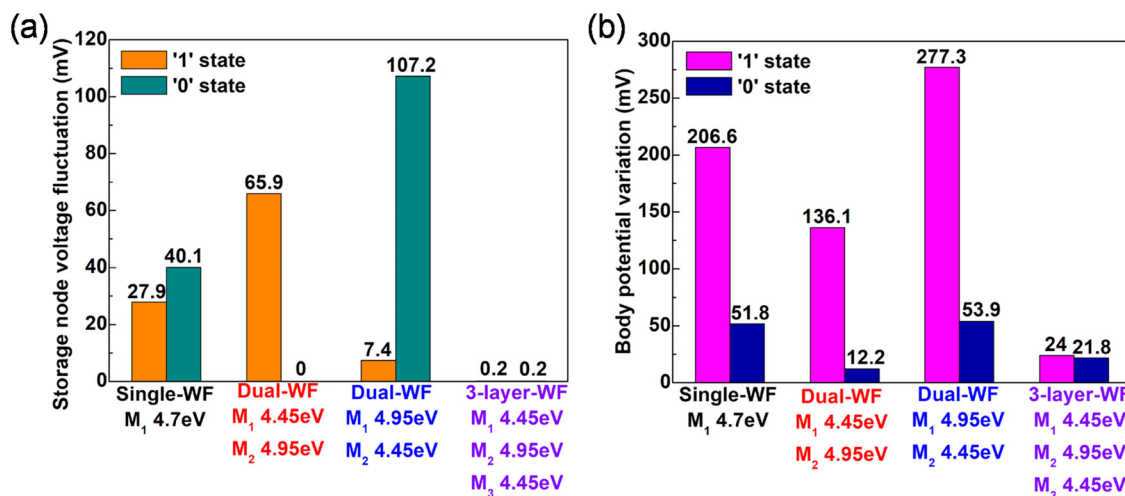


FIG. 4. Comparison of extracted dynamic retention metrics. (a) Storage node voltage (V_{SN}) fluctuation after ten BL toggling cycles under the “1” and “0” states. (b) Body potential variation during the BL-high period (t_1 – t_2).

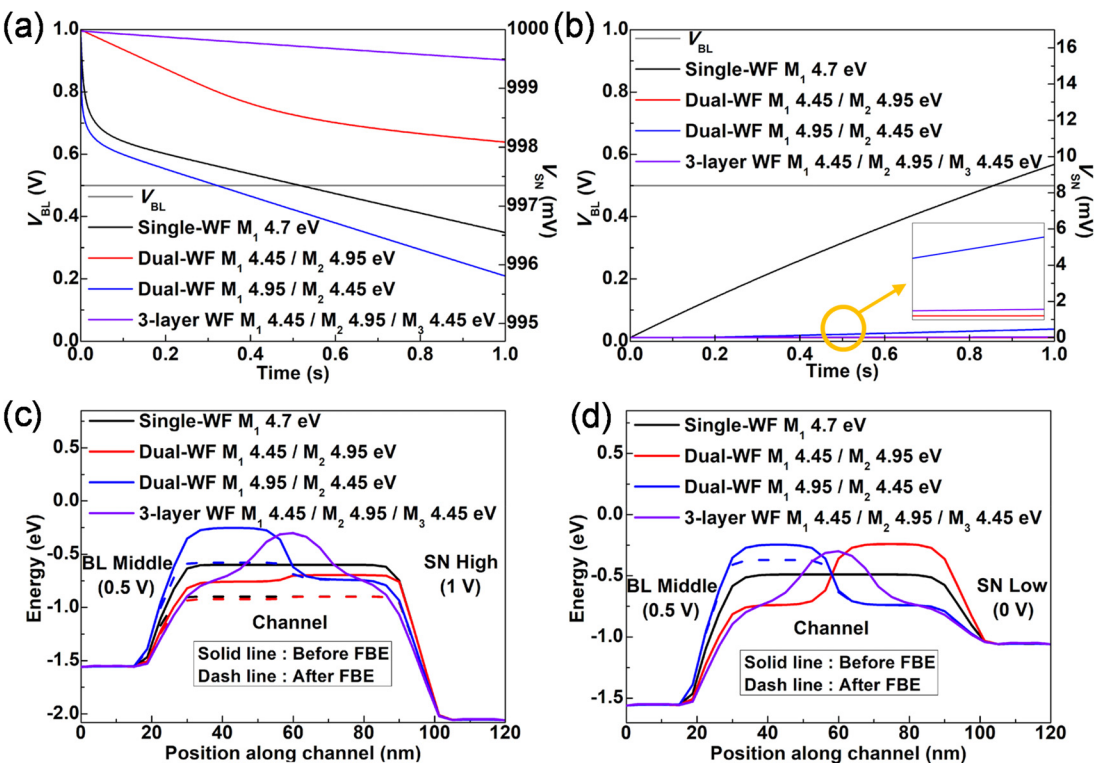


FIG. 5. Static retention characteristics of single-, dual-, and three-layer WF gate structures. (a) “1” state and (b) “0” state dynamic retention behavior showing the transient response of V_{SN} under $V_{BL}=0.5$ V. Valence-band energy profiles along the channel before and after hole accumulation of 0.1 s for the (c) “1” state and (d) “0” state.

respectively. Figures 5(c) and 5(d) depict the corresponding valence-band energy profiles along the vertical channel before and after hole accumulation.

When the SN stores a “1” state, GIDL occurs at the SN junction, gradually decreasing V_{SN} . As GIDL-induced hole injection increases the body potential, as shown in Fig. 5(c), the electric field at the SN junction is reduced, thereby suppressing further GIDL through FBE. Consequently, the V_{SN} decay rate decreases over time. The three-layer WF gate demonstrates the most stable behavior with minimal V_{SN} degradation, because the low WF at M3 and M1 suppresses GIDL at both the SN and BL junctions, while the high WF at M2 limits electron injection from the BL toward the SN.

When the SN stores a “0” state, electron injection from the SN toward the BL gradually increases V_{SN} . As holes injected into the body by GIDL at the BL junction raise the body potential, as shown in

Fig. 5(d), electron injection from the SN toward the BL is further enhanced, thereby promoting additional GIDL through FBE.

The dual-WF configuration with a high WF at M2 and low WF at M1 achieves the best retention by reducing the electric field at the BL junction and increasing the potential barrier between the SN and the body. The three-layer WF gate also exhibits good retention characteristics, whereas the single-WF gate shows the largest V_{SN} variation due to the absence of a high WF metal that maintains channel depletion. Therefore, the three-layer WF gate clearly demonstrates superior static retention for both the “1” and “0” states.

Table I summarizes representative FBE suppression technologies benchmarked against the proposed three-layer WF gate.^{14,17–19} The numbers in the table represent the percentage reduction in V_{SN} fluctuations achieved after applying each technology compared to before. Previous studies mainly focused on improving the “1” state retention,

TABLE I. Benchmark comparison of FBE suppression technologies.

Reference	14	17	18	19	This work
Engineering method	LDD	Multi-WL	Inner gate	Dual-WF	Three-layer WF
“1” state dynamic retention	78%	97.5%	~25%	85%	99.3%
“0” state dynamic retention	N/R	N/R	N/R	N/R	99.5%
“1” state static retention	N/R	N/R	Worse	80.8%	85%
“0” state static retention	N/R	N/R	N/R	N/R	99.6%

with limited or no enhancement in the “0” state retention, leaving many entries marked as not reported (N/R). In contrast, the proposed design improves both dynamic and static retention for both the “1” and “0” states. Overall, the three-layer WF gate most effectively suppresses V_{SN} fluctuations caused by FBE while maintaining a single-WL architecture. It should be noted that this strategy requires the use of thin metal layers with distinct work functions in scaled devices.

This study presented a comprehensive investigation of the FBE in VCTs and introduced a three-layer WF gate structure designed to mitigate FBE-induced retention degradation during DRAM operations. Mixed-mode TCAD simulations were conducted to quantitatively analyze both dynamic and static retention characteristics in the “1” and “0” states. The proposed three-layer WF gate achieved balanced suppression of FBE across both states, minimizing V_{SN} variations under all retention conditions, unlike previous engineering approaches that targeted only the “1” state. The proposed structure can be fabricated through sequential metal deposition without additional lithography, enabling scalable 4F² VCT DRAM technology with suppressed FBE.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Seung-Yoon Lee and Ki-Nam Park contributed equally to this work.

Seung-Yoon Lee: Data curation (lead); Investigation (lead); Methodology (lead); Validation (lead); Visualization (lead); Writing – original draft (lead). **Ki-Nam Park:** Methodology (supporting); Software (supporting). **Sihyun Kim:** Funding acquisition (equal); Resources (equal); Software (equal); Supervision (equal). **Joon-Kyu Han:** Conceptualization (equal); Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

REFERENCES

- ¹M. White, J. Qin, and J. B. Bernstein, in *Annual Reliability and Maintainability Symposium* (IEEE, 2021), p. 1.
- ²S. K. Gautam, A. Kumar, and S. K. Manhas, *IEEE Electron Device Lett.* **39**, 1286 (2018).
- ³Y. J. Yoon, J. H. Seo, S. Cho, J.-H. Lee, and I. M. Kang, *Appl. Phys. Lett.* **114**, 183503 (2019).
- ⁴S. Shiratake, in *IEEE International Memory Workshop (IMW)* (IEEE, 2020), p. 1.
- ⁵W. Mueller, M. Trentzsch, U. Meier, R. Rolfes, T. Schulz, W. Steinhögl, M. Kammler, and S. Buechner, in *IEEE International Electron Devices Meeting* (IEEE, 2005), p. 339.
- ⁶S. E. Kim, J. Y. Sung, J. D. Jeon, S. Y. Jang, H. M. Lee, S. M. Moon, S. W. Lee, J. G. Kang, H. J. Lim, and H.-S. Jung, *Adv. Mater. Technol.* **8**, 2200878 (2023).
- ⁷IEEE, *IEEE International Roadmap for Devices and Systems* (IEEE, 2022).
- ⁸C. J. Radens, S. Kudelka, L. Nesbit, R. Malik, and J. A. Mandelman, in *IEEE International Electron Devices Meeting* (IEEE, 2000), p. 349.
- ⁹A. Spessot and H. Oh, *IEEE Trans. Electron Devices* **67**, 1382 (2020).
- ¹⁰H. Chung, K. Park, Y. Lee, D. Lee, J. Kim, M. Ryu, Y. Cho, H. Kim, and H. Shin, in *European Solid-State Electronics Research Conference (ESSDERC)* (IEEE, 2011), p. 211.
- ¹¹J.-M. Yoon, J. Kim, Y. Lee, H. Chung, Y. Cho, H. Kim, and H. Shin, in *Device Research Conference* (IEEE, 2006), p. 259.
- ¹²Y. Cho, K. Park, H. Chung, J.-M. Yoon, J. Kim, and H. Shin, *IEEE Trans. Electron Devices* **65**, 3237 (2018).
- ¹³G. Giusi, *Solid-State Electron.* **95**, 1–7 (2014).
- ¹⁴J. Deng, X. Wang, Y. Ou, Z. Bai, T. Wang, X. Liu, M. Liu, Q. Hu, X. Wang, G. Wang, and C. Zhao, *IEEE Trans. Electron Devices* **72**, 199 (2025).
- ¹⁵J. Chen, F. Assaderaghi, P.-K. Ko, and C. Hu, *IEEE Electron Device Lett.* **13**, 572 (1992).
- ¹⁶S. Hong, in *IEEE International Electron Devices Meeting* (IEEE, 2010), p. 12.4.1.
- ¹⁷W.-C. Chen, J. Lee, C.-Y. Chou, T.-H. Wang, S.-H. Lin, Y.-S. Huang, M.-F. Tsai, P.-W. Hsu, C.-H. Chang, and C. Wu, in *IEEE International Electron Devices Meeting (IEDM)* (IEEE, 2024), p. 1.
- ¹⁸K. K. Min, S. Hwang, J.-H. Lee, and B.-G. Park, *IEEE Trans. Electron Devices* **67**, 944 (2020).
- ¹⁹I. Ahmad Khan, S. K. Manhas, A. Kumar, and M. Pakala, *IEEE Trans. Electron Devices* **71**, 7412 (2024).
- ²⁰D.-S. Park, Y.-J. Kim, M. Ryu, Y.-S. Lee, J.-H. Choi, H. Chung, Y. Cho, and H. Shin, *IEEE Access* **12**, 63049 (2024).
- ²¹Synopsys Inc., *Sentaurus TCAD User Guide* (Synopsys Inc., 2023).
- ²²D.-I. Moon, J. Park, S. Kim, Y. Hwang, J. Lee, and Y. Choi, in *IEEE International Electron Devices Meeting (IEDM)* (IEEE, 2011), p. 24.6.1.